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Advancements in digital twin-enhanced health monitoring and condition assessment of cable-supported bridges

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ABSTRACT

Cable-supported bridges are critical civil infrastructure. With increasing traffic volume and extreme weather conditions, these bridges face threats to their functional and structural safety during their service life, necessitating timely health monitoring and condition assessment. In recent years, digital twin (DT) has achieved significant progress in bridge monitoring and assessment, effectively addressing the shortcomings of traditional methods. However, current bridge DTs, especially those for cable-supported bridges, still face a series of issues, such as ambiguous concepts and lack of unified standards. To enrich the discussion on this topic, this paper explores the current research trends and frameworks of bridge DT through bibliometric analysis and a comprehensive review of the latest advancements, providing a systematic overview of relevant research on health monitoring and condition assessment of cable-supported bridges based on DT. Firstly, the origin, definition, and characteristics of DT are introduced. Secondly, a systematic literature review is employed to statistically analyze the current research status and discuss the construction methods of the DT system. Subsequently, a novel three-layer framework for DT is proposed, which can be applied to health monitoring and condition assessment of cable-supported bridges. Finally, the challenges and future trends of DT in this field are discussed. The challenges consist of seven aspects, which are data management, modeling, security and privacy, service applications, intelligence, integration, and standardization. Future DT development will focus on advancing digitization, integration, and intelligence, while deepening theoretical research and practical applications. The establishment of a reliable operational condition assessment framework is also an important trend.

1. Introduction

Cable-supported bridges, as crucial transportation infrastructure, face significant threats to their functional and structural safety over their service life due to material degradation and various other influencing factors, including traffic overloads, vehicle or ship collisions, fatigue cracking, steel corrosion, fires, typhoons, earthquakes, and especially the ever-increasing traffic demands and changing weather conditions [1]. Garg et al. [2] analyzed the failure causes of 2010 bridges from 1977 to 2017 and found that natural disasters accounted for 80.30 %, material degradation for 10.10 %, traffic overloads for 3.28 %, and human-induced disasters for 2.19 %. In reality, there are numerous specific cases of bridge damage. For example, the Humber Bridge in the UK, which experienced excessive wear of its bearings due to extreme loads [3], the Martinus Nijhoff Cable-Stayed Bridge in the Netherlands, which showed fatigue damage much earlier than expected during its design service life [4], and the Jiangyin Suspension Bridge in China, which suffered from the failure of expansion joints due to traffic loads [5]. Therefore, long-term and effective health monitoring and condition assessment of cable-supported bridges throughout their service life are crucial. Traditional monitoring and assessment methods for cable-supported bridges have met certain demands but still face challenges or issues [6–8].

Currently, digital twin (DT) has become a research hotspot [9–11] due to its unique advantages and is widely used for monitoring, predictive maintenance, and analysis of various physical assets. DT can be considered as a virtual model that represents an object throughout its lifecycle, maintaining its state, properties, and operational dynamics. Compared to traditional models, DT can provide richer information about structural conditions and, by integrating with the latest

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2352-0124/© 2025 Institution of Structural Engineers. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

discoveries in physical modeling and advanced algorithms, can offer near-real-time representations of operational bridge systems. Therefore, DT can significantly enhance the effectiveness of bridge health monitoring and condition assessment [12].

To enrich the research on DT in health monitoring and condition assessment of cable-supported bridges, this paper systematically reviews the research and applications of DT in this field, aiming to provide a roadmap for scholars and practitioners and lay the foundation for future discussions and research. The detailed objectives of this paper are as follows:

(1) Describe cable-supported bridges and their traditional monitoring and assessment technologies.
(2) Introduce the origin, definition, and characteristics of DT, and compare DT with similar concepts.
(3) Analyze the current research status of health monitoring and condition assessment of cable-supported bridges based on DT and discuss the methods for constructing DT system based on existing research.
(4) Propose a three-layer framework of DT suitable for the field of health monitoring and condition assessment of cable-supported bridges by analyzing existing research.
(5) Discuss the challenges and future development trends of DT in health monitoring and condition assessment of cable-supported bridges.

The remainder of this paper is organized as follows. Section 2 provides an overview of the background of cable-supported bridges, health monitoring and condition assessment, and DT. Section 3 describes the research methods followed in this paper and presents a statistical analysis of the selected literature. Section 4 discusses the construction of DT system. Section 5 introduces practical application cases of DT in health monitoring and condition assessment of cable-supported bridges. Section 6 presents and introduces a novel three-layer framework of DT for health monitoring and condition assessment of cable-supported bridges. Section 7 discusses and analyzes the challenges and future trends of DT in health monitoring and condition assessment of cable-supported bridges. Section 8 outlines the conclusions of this paper.

2. Background

2.1. Cable-supported bridges

Cable-supported bridges, primarily consisting of suspension bridges and cable-stayed bridges, represent one of the most significant investments in the national transportation network. The main components include towers, main cable/stay cables, and the deck. Fig. 1 illustrates schematic diagrams of both suspension bridge and cable-stayed bridge. Although cable-supported bridges account for only about 1.5 % of all bridges, they are an effective structural form for large spans and intricate environments [13–15]. In bridge categories with spans exceeding 200 m, cable-supported bridges account for over 90 % [16]. With a service life exceeding 100 years, cable-supported bridges may undergo material aging, traffic loads [17], hurricanes or typhoons [14], earthquakes [18], fires [16], etc., during their lifespan, which can significantly impact the functionality and structural safety of the bridges.

2.2. Methods of health monitoring and condition assessment

Ensuring the safety and uninterrupted operation of cable-supported bridges is a key concern for stakeholders. Health monitoring and condition assessment of these bridges are essential processes for addressing the aforementioned issues.

There are various traditional methods for health monitoring and condition assessment of cable-supported bridges. The most basic method, and one adopted by most countries, is visual inspection for bridge condition assessment. However, visual inspection involves extensive manual labor and is prone to neglecting defects [6,8]. Bridge testing can inspect the overall structure or major components of a bridge, providing valuable data support for bridge monitoring and assessment. But this method typically requires bridge closures, which can have certain traffic and economic impacts. Structural Health Monitoring (SHM) is described as the application of sensing technology and analysis of structural characteristics to identify structural damage or degradation [19]. The development of SHM has facilitated the advancement of health monitoring and condition assessment of cable-supported bridges [20–25], enabling real-time diagnosis of the health status of both local components and the overall structure of

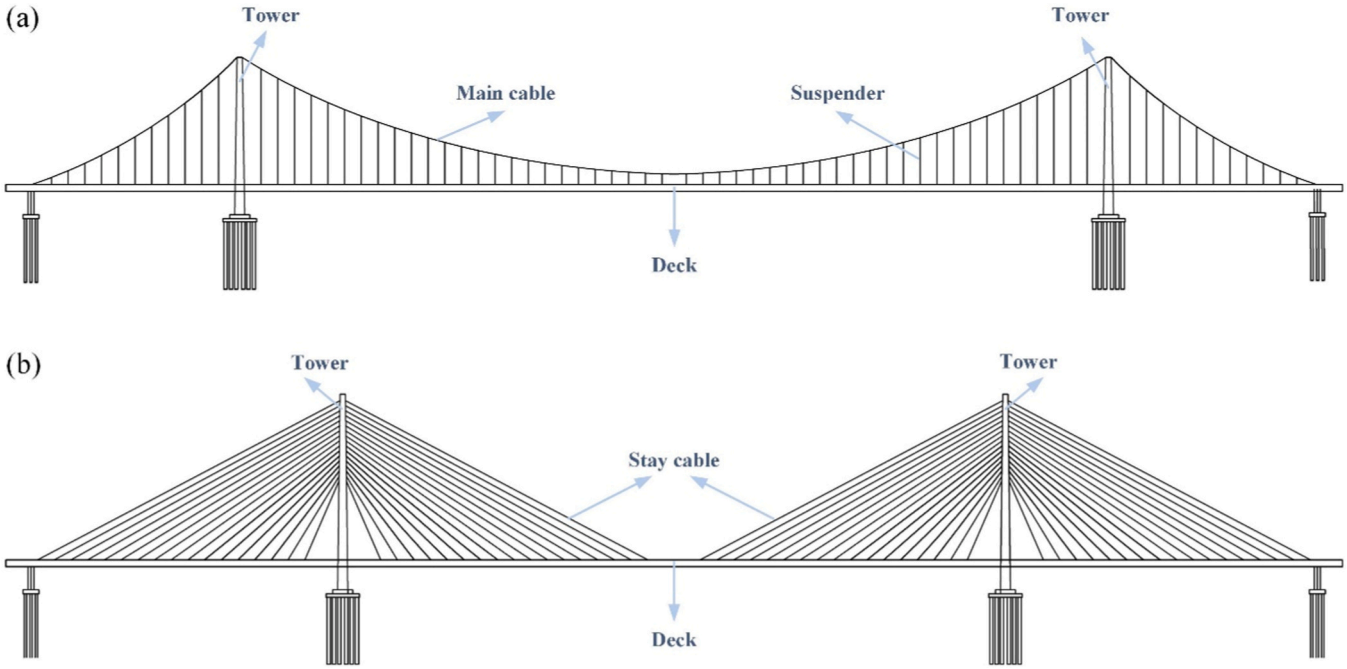


Fig. 1. Schematic diagram of the main types of cable-stayed bridges. (a) Suspension bridge, and (b) Cable-stayed bridge.

bridges [26]. However, SHM still faces challenges such as poor interoperability, data loss, low integration, low query efficiency and inability of sensor networks to comprehensively monitor the actual structural response [24].

Building Information Modeling (BIM) creates a virtual three-dimensional model and utilizes digital technology to provide a comprehensive, reality-aligned information database for construction projects [27]. BIM plays a crucial role in promoting digitization, visualization, and data exchange throughout the entire lifecycle of bridges [28]. However, it faces challenges such as lack of unified standards [29,30], poor data interoperability [31], high hardware requirements for model operation [32], and high technical barriers [33–36]. In recent years, data-driven technologies have been applied to the monitoring and assessment of cable-supported bridges [14,37–39], using data as the center for decision-making and action to achieve prediction and assessment of bridge performance. However, this method has high requirements for data quantity and quality and may also present risks of overfitting and poor interpretability [14].

2.3. Digital twin

2.3.1. Origin and definition of digital twin

In recent years, DT has been widely used in various fields [40] for the monitoring, analysis, and predictive maintenance of numerous physical assets, including buildings [41–43], factories, ships, cranes, aircraft, cities [44], and railways [45,46]. DT allows for personalized descriptions of physical assets in the form of computational models or parameters of interest, which evolve over time and maintain synchronization with the physical entities via data collection devices.

The origin of the DT can be traced back to 2002, when Grieves [47,48] proposed a conceptual model in a speech and used it in a course at the University of Michigan in 2003. Although the term "Digital Twin" did not appear in the model, it already encompassed all the essential elements of a DT, including real space, virtual space, bidirectional connection between real space and virtual space via data or information flow [49]. Therefore, it is widely recognized as the earliest introduction of the concept or idea of DT [50]. In 2010, the term "Digital Twin" was first coined by the National Aeronautics and Space Administration (NASA) [51], along with its formal definition.

Although DT has gained increasing popularity in academic research and industry practice, there is no universally accepted definition for it. Scholars in different fields have proposed various interpretations and definitions of DT. By synthesizing these previous definitions, this paper proposes the following definition of DT in the context of health monitoring and condition assessment of cable-supported bridges: a high-fidelity digital virtual representation of a cable-supported bridge, integrated with multiple technologies, capable of real-time data interaction with the physical entity and autonomous model updating, providing decision support and critical services for the health monitoring and condition assessment of the physical entity.

2.3.2. Digital twin characteristics

Fig. 2 illustrates the DT framework, which comprises three essential elements [47,52]: (1) the physical entity in the physical environment; (2) the DT in the virtual environment; and (3) the bidirectional connection between the physical entity and the DT, represented by arrows in the figure, with arrow directions indicating the direction of the connection. The "Collect" connection in Fig. 2 denotes the process of collecting data from the physical entity. Due to the vast amount of collected data and the possibility of some data being invalid or having indistinct features, data processing is necessary, corresponding to the "Process" connection. The "Feedback" connection indicates that the DT model can provide reverse feedback on the data after its establishment. The "Service" connection represents various services offered by the DT, such as simulation, decision-making, control, health monitoring, condition assessment, and data-driven operations of the physical entity, aimed at enhancing system convenience, reliability, and productivity.

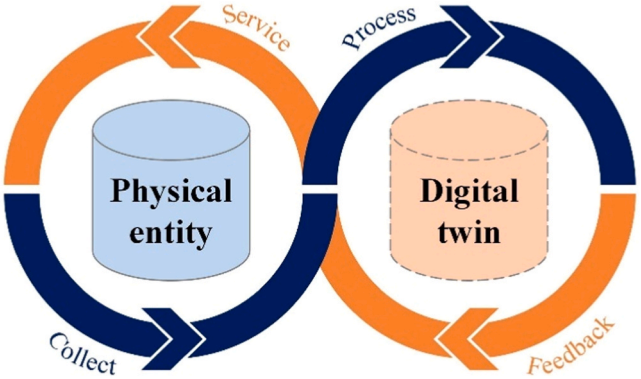


Fig. 2. Framework for digital twin.

DT possesses capabilities such as high integration and intelligence, real-time bidirectional data transmission [53], autonomous and rapid model updating [54], and the ability to reconstruct stress fields in areas lacking measurement data [24]. These capabilities effectively complement the deficiencies of traditional monitoring and assessment methods, assisting stakeholders in making reliable and swift decisions. Consequently, this enhances bridge performance, extends service life, and ultimately reduces maintenance and operational costs for the infrastructure.

2.3.3. Concept clarification

There are multiple concepts similar to the DT, such as Simulation (SIM), Building Information Modeling (BIM), Cyber-Physical Systems (CPS), and the Internet of Things (IoT). These similar concepts can easily be confused with the DT. Therefore, comparing these concepts with the DT can help to clarify the distinctions between them.

While the DT may seem similar to SIM, it is far more than that. A DT is a real-time, high-fidelity depiction of the operational dynamics of its corresponding physical entity. Operational dynamics are a crucial aspect of DT because its behavior is grounded in near-real-time data sourced from the actual physical entity. SIM emphasizes potential occurrences in the real world, rather than what is currently happening. A DT can serve multiple purposes such as monitoring, control, assessment, diagnosis, and prediction, extending beyond mere simulation.

BIM serves as a digital depiction of the physical and functional attributes of a building object [55]. Although both BIM and DT rely on virtual models, BIM emphasizes the existing physical entity it corresponds to, while DT has the capability to depict something that is nonexistent or yet to be built. Additionally, DT must reflect the current state of the physical entity in a timely manner, which is not a requirement for BIM [56].

A CPS is a multidimensional complex system that integrates computation, networking, and physical environments. Although both CPS and DT highlight the existence of a physical entity and involve timely data transmission between physical entity and virtual model, a DT necessitates a virtual model, which is not a necessity for a CPS (as CPS focuses more on the "cyber" aspect). Furthermore, a DT mandates a twinning relationship between the virtual model and the physical entity, which is not a requirement for CPS [56].

IOT refers to the connection between networks of physical assets, enabling data to flow among them. The secure deployment of communication protocols, the Internet, and computer networks facilitates this connection. Nevertheless, while it encompasses connectivity, IOT does not encompass the notion of a virtual model within the cyberspace [57].

The distinctions among DT, SIM, BIM, CPS, and IOT are presented in Table 1. Within the table, "× " indicates that the element is optional, whereas "√" indicates that the element is required.

Table 1
Distinctions among DT, SIM, BIM, CPS and IOT.

Elements	DT	SIM	BIM	CPS	IOT
Physical entity	√	×	×	√	√
Virtual model	√	√	√	×	×
Connections between physical and virtual	√	×	×	√	×
Twin relationship between physical and virtual	√	×	×	×	×

3. Research methodology and bibliometric analysis

3.1. Research methodology

The research method employed in this paper is Systematic Literature Review (SLR), which is a systematic and comprehensive approach to selecting, evaluating, and interpreting existing research within a specific field. As such, SLR can synthesize evidence and identify the current research status, challenges, and opportunities based on relevant studies [58]. The literature review strategy is illustrated in Fig. 3.

Selecting appropriate search terms as the basis for database searches is crucial given the scope and theme of this paper. "Digital twin" is the technology of primary focus in this paper, and hence it is chosen as the primary search term. It should be noted that although there are terms closely related to the concept of DT, such as Digital model, Simulation model, and Surrogate model, they are not equivalent to DT [59]. Therefore, only "Digital twin" is selected as the primary search term. There are diverse types of bridges, such as beam bridges and arch bridges, but this paper primarily focuses on cable-supported bridges, which mainly include suspension bridges and cable-stayed bridges. Therefore, the following two sets of terms are set as search terms, with the search fields being Title, Abstract, Keywords or indexing. Considering that some studies are applicable to all types of bridges, including cable-supported bridges, supplementary searches are conducted with additional phrases "bridge and health monitoring" and "bridge and condition assessment" added to the second set of search terms.

- Search Term 1: "Digital twin".
- Search Term 2: "cable-supported bridge", "suspension bridge", "cable-stayed bridge", "bridge and health monitoring", "bridge and condition assessment".

To comprehensively collect existing research, data was gathered from four renowned databases: Scopus, ScienceDirect, Web of Science (WOS), and IEEE Xplore. All journal articles or conference proceedings published in English were included, and considering that 2013 is regarded as the nascent stage of DT [60], the search period spanned from January 1, 2013, to August 5, 2024.

Based on the aforementioned criteria, a total of 416 records were retrieved from the four databases. Subsequently, these 416 records were screened according to the following three principles: (1) elimination of duplicate records; (2) exclusion of records that were inconsistent with the topic discussed in this paper; (3) selection of papers that included specific application cases related to the topic. After this screening process, 19 papers [18,24,49,61–76] were ultimately obtained. These papers will be discussed and analyzed in the subsequent sections of this paper. The number of finally obtained papers indicates that the application of DT in health monitoring and condition assessment of cable-supported bridges is currently in its initial stage, and related research is quite limited.

3.2. Bibliometric analysis

This subsection presents statistical data related to the selected literature. Such an analysis can provide important information. For example, identifying the research status of DT in the field of health monitoring and condition assessment of cable-supported bridges can assist scholars and practitioners interested in this field to identify opportunities for collaboration and further research.

Fig. 4(a) illustrates the annual distribution of the selected literature, indicating that while the overall number of papers per year is quite limited, there has been a gradual increase in research attention in this field. The number of papers published in the last three years dominates, with a particularly rapid growth in 2024, considering that this study only includes research conducted before August 5, 2024. As shown in Fig. 4(b), journal articles account for 73.7 % of the total publications, while conference proceedings make up 26.3 %. Furthermore, Fig. 4(c) presents the distribution of journal activity in this research field, revealing that the selected literature is most frequently published in the journals "Automation in Construction (Autom. Constr.)" and "Sensors", both accounting for 14.3 % of the total publications. To identify the countries with active DT applications in this research field, Fig. 4(d) categorizes the papers based on the country of the first affiliated institution. The results indicate that the two countries with the highest number of published papers in this field are China and the United Kingdom. Additionally, the statistics reveal a lack of collaboration among countries in this research area, highlighting the need for enhanced international cooperation to facilitate global knowledge exchange.

The VOSviewer software [77] possesses robust capabilities in generating, visualizing, and utilizing bibliometric networks [78,79]. Therefore, this software was employed to conduct a correlation analysis of the keywords in the selected literature, as illustrated in Fig. 5. The network in the figure is generated using points, lines, and colors. The points represent clusters of keyword co-occurrences, the lines indicate the established relationships between them, and the colors are used to identify these relationships based on themes [80].

Specifically, looking at the content presented in Fig. 5, the three largest points are "digital twin," "structural health monitoring," and "cable-stayed bridges," which are precisely the main topics of this paper. The remaining points are all directly or indirectly related to these three points. Focusing on the technology-related points in the figure, we can see terms such as "finite element modeling," "internet of things," "machine learning," "e-learning," "data integration," and "mapping", which demonstrate the advantages of DT in integrating multiple technologies and leveraging cutting-edge technologies.

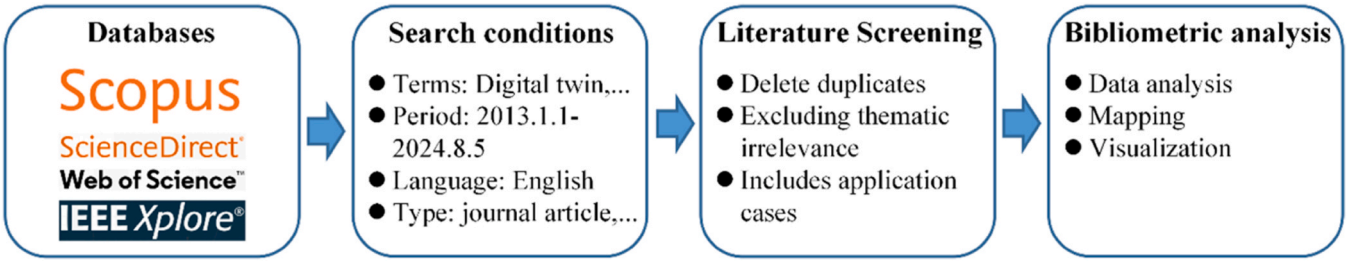


Fig. 3. Literature review strategy.

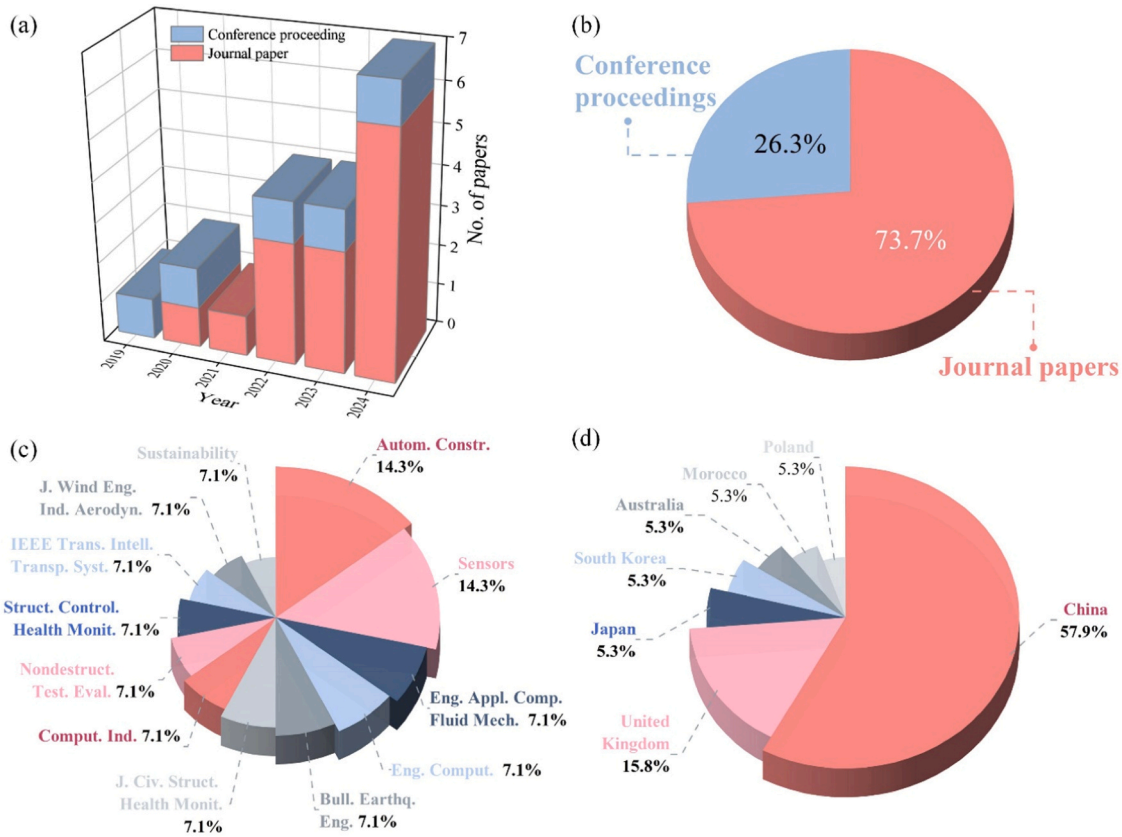


Fig. 4. Literature information statistics. (a) Annual distribution of publications and document types, (b) distribution of document types, (c) distribution of journal publishers, and (d) distribution of countries to which papers belong.

4. Methodologies and technologies in digital twin system

4.1. Data collection methods

Data is a crucial component of the DT system, including data generated by both the physical entity and its DT. This subsection mainly introduces the data collection methods for the physical entity, which play a key role in the establishment of DT model and real-time synchronization between physical entity and its DT.

4.1.1. Historical documentation

Historical documentation for bridges includes design information, construction documents, historical records, inspection reports, regulatory standards, and expert decision-making information. These data can provide support for the establishment of a DT model, such as the dimensions, material properties, and age of the physical entity.

4.1.2. Contact sensors

Deploying contact sensors is a commonly used method in bridge health monitoring and condition assessment to obtain bridge performance indicators. Common contact sensors include stress and strain sensors, displacement sensors, vibration sensors, temperature sensors, inclination sensors, anchor cable dynamometer, weigh-in-motion (WIM) sensors, etc. Installing multiple sensors at different locations of the structure according to requirements can obtain data information under normal operating conditions or extreme weather events, such as stress and strain, displacement, acceleration, velocity, temperature, inclination angle, cable force, and traffic load.

4.1.3. Laser and radar

Laser scanner and Ground-Penetrating Radar (GPR), as non-contact and non-destructive technologies, have been widely recognized and applied.

Laser scanner can quickly and accurately collect point cloud data from the surface of the physical entity. Based on the position of the laser sensor during data collection, it can be classified into three types: ground, aerial, and mobile. Terrestrial Laser Scanner (TLS) is capable of capturing intricate details of the entire scene along with its color attributes. Due to advantages such as sub-millimeter accuracy and fast scanning speed, this technology has been used in numerous studies [81] to obtain bridge information. Mounting laser scanners on equipment such as cars and unmanned aerial vehicles (UAVs) makes scanning operations more flexible, with a broader scanning coverage and higher collection efficiency.

GPR stands as one of the most extensively researched non-contact technologies. It determines internal defects and reinforcement layouts of bridge decks by quantifying the attenuation of reflected waves [82,83], and also provides important data support for the construction of DT system.

4.1.4. Image technology

With the rapid development of UAV technology and photographic techniques, UAV photography has become a practical and efficient method for data acquisition [84,85]. Images captured by UAVs can be processed and used for the construction and updating of the DT model. Although UAV photography offers advantages such as rapid, precise, and extensive remote data collection, there are still certain issues, such as path planning, image matching quality, image orientation accuracy, and modeling geometric accuracy. Additionally, in practical applications, there are blind spots where UAVs cannot capture images, requiring manual image capture by humans for areas inaccessible to UAVs.

Infrared Thermography (IRT) emits infrared radiation towards the physical entity and uses images to convert the thermal distribution data

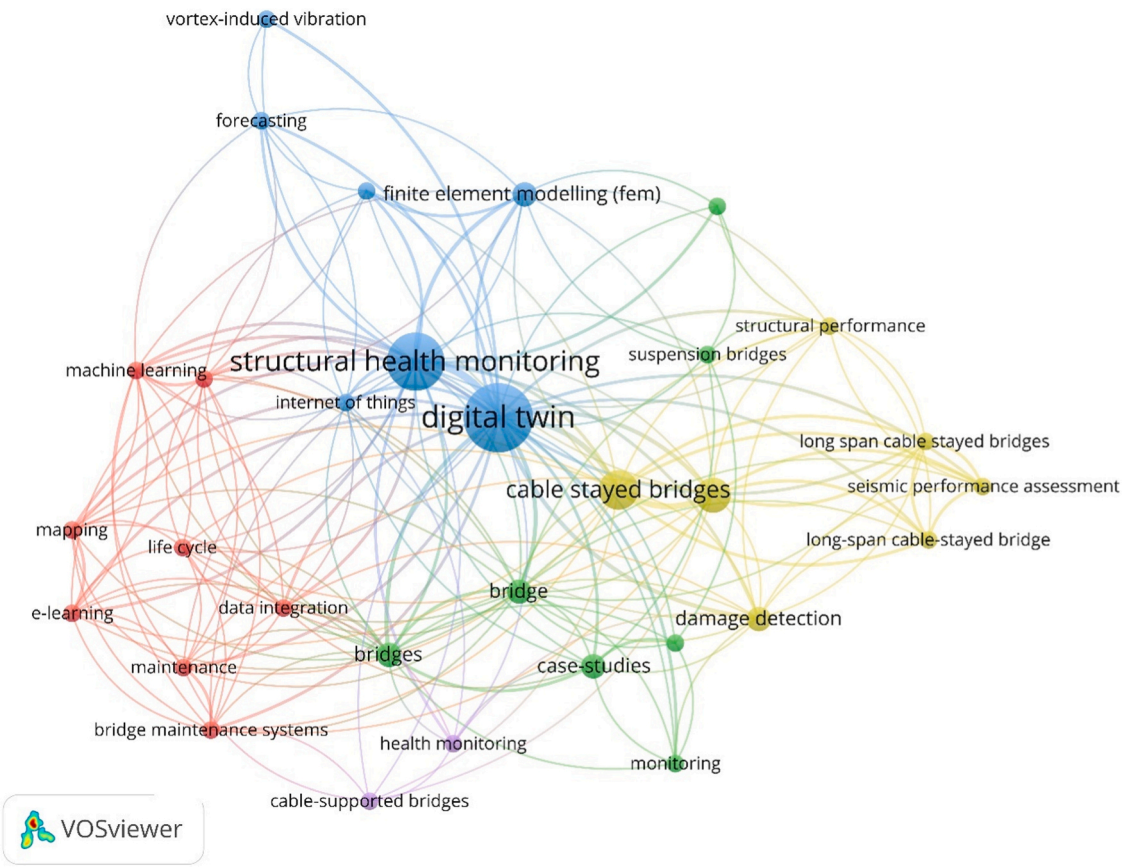


Fig. 5. Network of co-occurring keywords.

of the physical entity into visual representations. It can be used to obtain information such as the type and location of internal defects in bridges [86–89].

Hyperspectral imaging is capable of furnishing RGB information and surface composition data, relying on wavelength-specific responses, with a wavelength range from near-infrared to short-wave infrared [90]. It can be used to obtain information about the condition of concrete [91] and steel corrosion [92] in bridges.

4.1.5. *Multi-method integration*

SHM system is used for data collection and analysis of the physical entity, where the data collection method involves deploying devices (sensors) on the physical entity to monitor any changes. These devices can be a combination of contact and non-contact sensors.

4.2. *Data transmission methods*

Data transmission in DT is bidirectional, which provides a guarantee for real-time synchronization of information between DT and the physical entity. Moreover, data transmission must possess the capability to transfer data with satisfactory latency and cost [66], in order to facilitate the updating of the DT model and its feedback and services to the physical entity.

4.2.1. *Wired transmission*

Wired connections, such as fieldbuses and Ethernet, are among the most commonly used methods for data transmission. Due to their low cost and sufficient bandwidth, they can achieve fast and stable data transmission [93], and are therefore widely used for data transmission between DT and physical entity.

4.2.2. *Wireless transmission*

Wireless Sensor Networks (WSN) offer high energy efficiency and reliability in data transmission. Compared to wired transmission, they are more flexible and convenient for communication, making them suitable for monitoring bridges or components that are difficult to access via wired connections [94]. WSN is highly suitable for real-time data collection scenarios and holds a pivotal role in the construction of the DT system. Wireless communication technologies can be classified into short-range, medium-range, and long-range based on communication distance. Short-range wireless communication technologies include Wi-Fi, Bluetooth, and Zigbee. Medium-range wireless data transmission technologies include commercial cellular networks (such as 3 G, 4 G, and 5 G) and low-power wide-area networks (LPWAN, such as NB-IoT and LTE-M). Long-range wireless transmission technologies include LoRa, Sigfox, and Ingenu, which are ideal for bridge monitoring in remote areas with limited resources.

Advancements in wireless transmission technology have effectively facilitated the development of DT technology in the field of health monitoring and condition assessment of civil infrastructures, which has been extensively studied by scholars. Lynch [95] introduced the characteristics, applicable scenarios, and applications of wireless transmission technology, proposing a wireless active sensing unit design method. Using the Alamosa Canyon Bridge in New Mexico as an example, the accuracy and reliability of the wireless sensing units were verified. Zou et al. [96] designed an efficient multihop communication service, incorporating a flooding-based routing method and a theoretically optimal data collection protocol within the proposed protocols, achieving optimization of network performance and energy consumption. To address the issue of poor synchronous sensing during data collection, Fu et al. [97] proposed a time synchronization method suitable for wireless monitoring of civil infrastructures during emergencies, improving the efficiency and precision of time synchronization.

Additionally, Li et al. [98] proposed a synchronized sensing approach that incorporates nonlinear clock drift compensation to achieve efficient and accurate synchronous sensing. Bao et al. [99] introduced a compressive sensing-based method for recovering lost data to address the issue of data packet loss during wireless transmission, and verified its effectiveness through experiments and field tests.

4.3. Digital twin modeling methods

As the primary framework of DT, the generation of bridge's virtual model is a crucial and complex stage. The quality of the virtual model holds considerable significance in the practical application of the DT [100]. The following section outlines four modeling methods for DT.

4.3.1. Geometric-based method

Geometric modeling can be accomplished using 3D modeling software such as AutoCAD [71], Autodesk Civil 3D [49], Revit [70], Tekla Structures [67], and Solidworks. Additionally, commercial reverse engineering software can be employed to perform reverse modeling on point cloud data [101], which can originate from laser scanning or photogrammetry techniques, ultimately constructing a visual mapping of the geometric appearance of the physical entity.

4.3.2. Finite element method

Finite Element Method (FEM) modeling is a widely adopted modeling technique [102], with commonly used software such as Abaqus, MSC.Marc [63], Midas Gen [65,66], Ansys [73,75], etc. FEM employs discretization-derived mathematical approximations, and its complexity is positively correlated with the complexity of the physical entity. This results in FEM typically requiring excessive computational resources, making it difficult for the constructed virtual model to maintain real-time synchronization with the physical entity. Moreover, limitations in the size and time step of the virtual models make them unsuitable for real-time simulation environments in subsequent services [103,104]. Additionally, FEM modeling faces challenges in accurate visualization as it often relies on certain simplification methods.

4.3.3. Data-driven method

Data-driven modeling is based on data derivation and parameterization. Specifically, it employs artificial intelligence (AI) and machine learning (ML) techniques [105] to extract both implicit and explicit knowledge underlying specific phenomena of the physical entity from data. Data-driven models can reduce modeling workload and offer higher accuracy and broader adaptability. However, data-driven models have high requirements for the quantity and quality of data, and their black-box nature, due to opaque reasoning mechanisms and lack of physical consistency, poses certain obstacles to their use.

4.3.4. Multi-method integration

Combining multiple modeling methods can, to some extent, compensate for the deficiencies of each individual method. For example, Kang et al. [106] introduced a DT modeling method based on multimedia knowledge by combining the FEM and data-driven approaches. This method balances the advantages of both FEM and data-driven methods, facilitating better service for bridge health monitoring. Additionally, incorporating physical information (such as physical formulas, principles, etc.) into data-driven methods can enhance the interpretability of the model.

4.4. Services provided by the digital twin

The application purpose of DT is to provide services to physical entities, such as monitoring, prediction, simulation, decision-making, lifecycle management, and condition assessment [24,49], enhancing the system's reliability, convenience, and overall productivity.

Monitoring service is regarded as a key service for bridge DT, allowing real-time understanding of the performance and status of the physical bridge [107]. The prediction service aims to forecast the future behavior, health status, and structural lifespan of the physical bridge, providing a basis for proactive measures and enhancing reliability. The simulation service replicates real-world scenarios in a virtual environment with high fidelity for structural integrity analysis. The decision-making service integrates and analyzes data from monitoring, prediction, and simulation services to form accurate and effective decisions, while also facilitating strategic planning throughout the bridge's lifecycle [67]. Lifecycle management service can reduce long-term repair costs, optimizing operational costs, structural integrity, and safety [108]. Condition assessment service utilizes multi-source data fusion to evaluate the current health status and structural integrity of the bridge. It identifies structural degradation, potential damage, and critical areas requiring maintenance or repair.

4.5. Discussion on digital twin system implementation

This section outlines the methodologies for constructing a DT system tailored for health monitoring and condition assessment of cable-supported bridge. The data acquisition techniques in Section 4.1 emphasizes integrating diverse sources, including historical documentation, contact sensors and advanced non-contact methods such as laser scanning, radar, UAV imagery, and hyperspectral imaging. While each method offers unique advantages, their combined application enables comprehensive data capture encompassing geometric, environmental, and structural response parameters. However, challenges remain, including sensor durability, precision, UAV blind spots, and the complexity of managing heterogeneous, high-volume data.

Section 4.2 addresses data transmission, which is essential for real-time synchronization between the physical and virtual entities. Wired methods offer high reliability and bandwidth, while wireless solutions provide flexibility for remote and inaccessible components. Ensuring low latency, high reliability, and robust protocols remains paramount, especially given the substantial data volumes generated by continuous monitoring systems.

Section 4.3 reviews DT modeling techniques. Geometric modeling provides essential visual fidelity but inherently lacks the capability to simulate physical behavior. FEM offers valuable physics-based simulation capabilities, yet faces significant computational burdens and limitations in accurate visualization. Data-driven models promise efficiency and adaptability but demand high-quality, extensive datasets and suffer from interpretability issues due to their "black-box" nature. Thus, hybrid approaches, particularly combining FEM with data-driven techniques or incorporating physical knowledge into ML models, emerges as a highly promising solution to balance accuracy, computational efficiency, and interpretability.

Finally, Section 4.4 outlines the core services enabled by DT systems, including monitoring, prediction, simulation, decision-making, lifecycle management, and condition assessment. These services move beyond mere data visualization towards generating actionable insights and enabling proactive management. The effectiveness of these services is fundamentally dependent on the robustness and synergy of data acquisition, transmission, and modeling.

5. Digital twin application in health monitoring and condition assessment of cable-supported bridges

For cable-supported bridge managers and researchers interested in this field, understanding and mastering the practical application of DT in health monitoring and condition assessment of cable-supported bridges is crucial, as it provides significant value for the research, updating, and deployment of DT. This section provides a detailed review and analysis of the 19 selected literature to clarify the specific applications of DT in health monitoring and condition assessment of cable-supported bridges, as shown in Table 2. By observing the physical entities in Table 2, the

Table 2
Specific application of digital twins in the selected literature.

No.	Ref.	Physical entity	Digital twin system			
			Technologies			Services
			Data collection	Data transmission	Modeling	
1	[61]	Suspension bridge, South Korea	Historical documentation, Inspection	–	BIM	Monitoring, prediction
2	[49]	Mohammed VI Bridge, Morocco	Dynamic Laser Scanning	–	Autodesk Civil 3D, 3D GIS	Prediction
3	[62]	Experimental suspension bridge	Camera	–	–	Monitoring, condition assessment
4	[63]	Physical scaled model of Sutong Bridge, China	Test	–	MSC.Marc, MATLAB, Python	Simulation (MSC.Marc, MATLAB, Python), Condition assessment
5	[64]	Bridges group in Shanghai, China	Camera, LIDAR, WIM	Cloud server, 4 G	–	Monitoring, simulation (OpenSees), condition assessment
6	[66]	Clifton Suspension Bridge, UK	Historical documentation, contact sensor	Wired and wireless transmission	Midas Gen, NODLE FEM	Monitoring, prediction, simulation (Midas), condition assessment
7	[65]	Clifton Suspension Bridge, UK	Historical documentation, SHM	–	Midas Gen, NODLE FEM	Simulation (Midas), condition assessment
8	[24]	Yangtze River highway bridge, China	Test, Inspection, SHM, WIM	–	–	Prediction, condition assessment
9	[68]	Clifton Suspension Bridge, UK	Historical documentation, SHM	Wired and wireless transmission	Midas Gen, NODLE FEM	Monitoring, prediction, simulation (Midas), condition assessment
10	[67]	Werrington Bridge, Australia	TLS	–	Tekla Structures	Decision, condition assessment
11	[69]	Long-span cable-stayed railway-highway combined bridge in Anhui, China	SHM	–	Deep learning	Monitoring, condition assessment
12	[18]	Physical scaled model of Sutong Bridge, China	Test	–	MSC.Marc, MATLAB, Python	Prediction, lifecycle management, condition assessment
13	[72]	Huai River Highway Bridge, China	Contact sensor, laser displacement sensor	LTE-TDD, LTE-FDD, cloud platforms, websocket	3D modelling, FE model, surrogate model	Monitoring, simulation, condition assessment
14	[70]	Cable-stayed bridge on Poland's National Highway 75	SHM	Web server	BIM	Monitoring, prediction, simulation (AXIS VM, ATENA 3D), decision
15	[74]	Forth Road Bridge, UK	GeoSHM	–	FEM	Prediction, simulation (SAP2000)
16	[76]	Sutong Bridge, Runyang Bridge, China	Historical documentation, Inspection, SHM	–	ANSYS	Prediction, simulation (ANSYS), condition assessment
17	[75]	Xihoumen Bridge, China	SHM	–	ANSYS	Prediction, simulation (ANSYS)
18	[71]	Laboratory-scale single-tower cable-stayed bridge	Test	–	CAD、CAE	Monitoring, lifecycle management, condition assessment
19	[73]	Physical scaled model of Xiangshan Harbor Bridge, China	Test	–	ANSYS FLUENT	Prediction, simulation (ANSYS FLUENT)

Note: The full names of LTE-TDD, LTE-FDD and GeoSHM are long-term evolution time-division duplex, long-term evolution frequency division duplex and GNSS and earth observation for structural health monitoring, respectively.

studies in the table can be classified into two categories: those where the physical entity is the bridge itself being studied and those where the physical entity is a scaled-down model of the bridge being studied.

5.1. Real bridges

Shim et al. [61] introduced the concept of DT into the maintenance system of cable-supported bridges and integrated it with the monitoring system, enabling real-time monitoring and prediction of bridge behavior and performance. This was illustrated using a suspension bridge in Korea as an example. Sofia et al. [49] proposed an idea to integrate DT with geospatial data for predicting the future performance of the Mohammed VI Bridge in Morocco under different environmental conditions. Dan et al. [64] incorporated machine vision into a bridge traffic load monitoring system, proposed the DT for a group of bridges within a regional transportation infrastructure network, and validated it using a regional bridge group in Shanghai, China, achieving awareness of the working status and safety warnings for all bridges in the regional transportation network. Pregnolato et al. [66,68] and Voyiatzis et al. [65] proposed a five-step framework for developing DT and demonstrated its application using the Clifton Suspension Bridge in the UK as an example, aiming to achieve health monitoring and condition assessment of the bridge.

Yu et al. [24] proposed a DT for steel bridge decks of cable-stayed bridges that integrates monitoring data with numerical models. Taking into account the coupling effects of vehicle loads, welding residual stress, and temperature variations, this approach enables fatigue assessment and fatigue life prediction of steel bridge decks. Furthermore, the Yangtze River highway bridge in China is used as an example to illustrate the application. Mohammadi et al. [67], using the Werrington Bridge in Australia as an example, proposed a bridge management framework that integrates TLS-derived Bridge Information Modeling (BrIM) with Decision Support Systems (DSS) for decision-making and condition assessment of bridges. Zhao et al. [69] investigated DT modeling methods for distributed mapping from structural temperature fields to strain fields induced by temperature. They proposed a digital modeling approach based on deep learning and clustering theory for perceiving the redistribution of strain fields under non-uniform temperature fields. Using a long-span cable-stayed railway-highway combined bridge in Anhui, China as an example, they analyzed the characteristics of long-term time-series data of temperature and temperature-induced strain. Lai et al. [72] combined the advantages of finite element models and surrogate models, treating each finite element node as a virtual sensor, and proposed a non-destructive testing method based on DT. This approach achieves full-bridge condition monitoring while enhancing computational efficiency and accuracy. Subsequently, the effectiveness and feasibility of the approach were verified using the Huai River Highway Bridge in China as an example.

Fawad et al. [70] developed an immersive DT framework that integrates VR/AR technology, BIM, IoT, and a 3D game engine to support health monitoring and management of bridges, as well as predictive decision-making. The effectiveness of this framework was tested on a

cable-stayed bridge on Poland's National Highway 75. Liu et al. [74] proposed two DT dual-drive construction methods: a Bayesian neural network (BNN) model and a finite element model improved through multi-objective optimization (MOO) using genetic algorithms (GAs) and the response surface method (RSM). These methods aim to predict structural dynamic response and displacement under strong winds. The characteristics of these two methods were assessed using the Forth Road Bridge in the UK as an example. Zhang et al. [76] proposed a comprehensive bridge assessment system based on advanced sensor-guided structural health monitoring data and a probability-based DT. This system is used to evaluate the safety, functionality, and environmental conditions of long-span cable-supported bridges during their service life. The Sutong Bridge and Runyang Bridge in China were used as examples to introduce this assessment system. Zhang et al. [75] developed a DT for vortex-induced vibrations of long-span suspension bridges by combining SHM data with physical models to achieve real-time simulation and prediction of vortex-induced vibrations in bridges. The accuracy of this model was verified and evaluated using the Xihoumen Bridge in China as an example.

5.2. Physical scaled model

Shao et al. [62] applied holographic vision sensors and their algorithms to full-field non-contact displacement and vibration measurements in computer vision. They validated this approach using a test suspension bridge, achieving holographic monitoring of the bridge structure's geometry. This work lays the foundation for DT and structural condition assessment of large-scale structures. Lin et al. [18,63] introduced a DT-based approach to evaluate the seismic performance and collapse vulnerability of long-span cable-stayed bridges under strong earthquakes. They verified the feasibility and accuracy of the proposed approach using a physical scaled model of the Sutong Bridge. Guo and Fang [71] combined the advantages of model-driven and data-driven methods to propose a DT modeling approach for the full lifecycle of cable-stayed bridges. They validated the effectiveness of this method using a laboratory-scale single-tower cable-stayed bridge. Li et al. [73] established a DT model by combining computational fluid dynamics (CFD) simulations with wind tunnel tests to accurately and efficiently predict vortex-induced vibration responses of bridge decks. They conducted research and validation using a physical scaled model of the Xiangshan Harbor Bridge as an example.

5.3. Discussion on digital twin applications

This section systematically reviews 19 practical applications of DT in health monitoring and condition assessment of cable-supported bridges. These applications are categorized into real-world structures (Section 5.1) and physical scaled models (Section 5.2). The case studies collectively demonstrate the versatility of DT frameworks in integrating diverse technologies. These technologies include laser scanning, IoT sensors, BIM, machine learning, and finite element modeling. They enable real-time monitoring, predictive analytics, and condition evaluation. Key observations are as follows.

(1) While all studies leverage DT principles, methodologies vary significantly in data acquisition (e.g., contact sensors vs. UAV-based imaging), modeling techniques (FEM vs. data-driven surrogates), and service outputs (fatigue prediction vs. seismic assessment). This reflects the context-specific adaptability of DTs but highlights a lack of standardized workflows.
(2) Applications to full-scale bridges (e.g., Clifton Suspension Bridge, Yangtze River Bridge) validate DT robustness under operational conditions, yet computational demands for high-fidelity models remain challenging. In contrast, physical models (e.g., Sutong Bridge prototype) facilitate controlled validation but may not fully capture complex environmental interactions.
(3) Few studies fully exploit bidirectional DT capabilities; most emphasize data-to-model synchronization but seldom use model-to-physical feedback for autonomous control or adaptive maintenance. Additionally, interoperability between legacy monitoring systems and emerging DT architectures requires further attention.

These findings underscore DT's transformative potential in enhancing bridge resilience but also reveal critical research needs, i.e., standardization of modular DT frameworks, validation of long-term autonomy under extreme events, and scalable solutions for legacy infrastructure integration.

6. The three-layer digital twin framework

Through compiling and analyzing existing research, this paper proposes a three-layer framework for DT, as shown in Fig. 6. This framework is applicable to the field of health monitoring and condition assessment of cable-supported bridges, clearly presenting the content and processes of the DT system. It provides a reference for the research and application of DT in cable-supported bridges. Furthermore, the DT constructed within this framework incorporates the advantages of both the finite element model and the data-driven model, enabling it to better serve the physical entity. This framework consists of three layers: the data layer, the model layer, and the service layer.

The data layer encompasses data collection, transmission, and processing, serving as the most fundamental layer within the three-layer framework. Cable-supported bridges requiring health monitoring and condition assessment typically possess historical documentation, including design, construction, inspection, and decision-making materials. Furthermore, a SHM system is generally designed and installed on the bridge during its construction phase to facilitate subsequent monitoring activities during operation. Therefore, data collection can rely on both historical documentation and the SHM system. To obtain as much bridge data as possible, instruments utilizing technologies such as cameras, UVAs, lasers, and radars can be deployed on cable-supported bridges as supplementary devices, in addition to traditional sensors. Data transmission employs a combination of wired and wireless methods, tailored to the actual conditions of the bridge, to achieve the most suitable data transmission arrangement. Data processing begins with pre-processing large volumes of data, including de-noising, outlier detection, cleaning, and normalization. Subsequently, further processing is conducted based on the detailed requirements of operation managers. Finally, the processed data is stored in a database for future use.

The model layer encompasses model construction and model updating. The DT model is constructed using a combined approach of finite element modeling and data-driven modeling, harnessing the advantages of both methods. For cable-supported bridges with ample historical documentation, such as sizes, positions, and material properties, the bridge model can be established using finite element software Abaqus [109]. For cable-supported bridges with limited historical documentation, point cloud and image data from the database can be utilized. Specifically, point cloud data is first converted into a mesh model using surface reconstruction algorithms, then transformed into the INP file format acceptable by Abaqus to obtain the bridge's Abaqus model. Subsequently, feature extraction is performed on the image data to obtain key data, which can be used for setting boundary conditions and subsequent simulation analysis in the bridge's Abaqus model. This finite element model offers advantages such as good visualization and ease of simulation.

In addition to the finite element method, data-driven modeling is also employed as a complement to the finite element model. It utilizes Python software [110] and some cutting-edge, efficient algorithms and programs to establish a data-driven model of the cable-supported bridge. When dealing with complex modeling situations or when the service provided by the finite element model is unsatisfactory, the data-driven

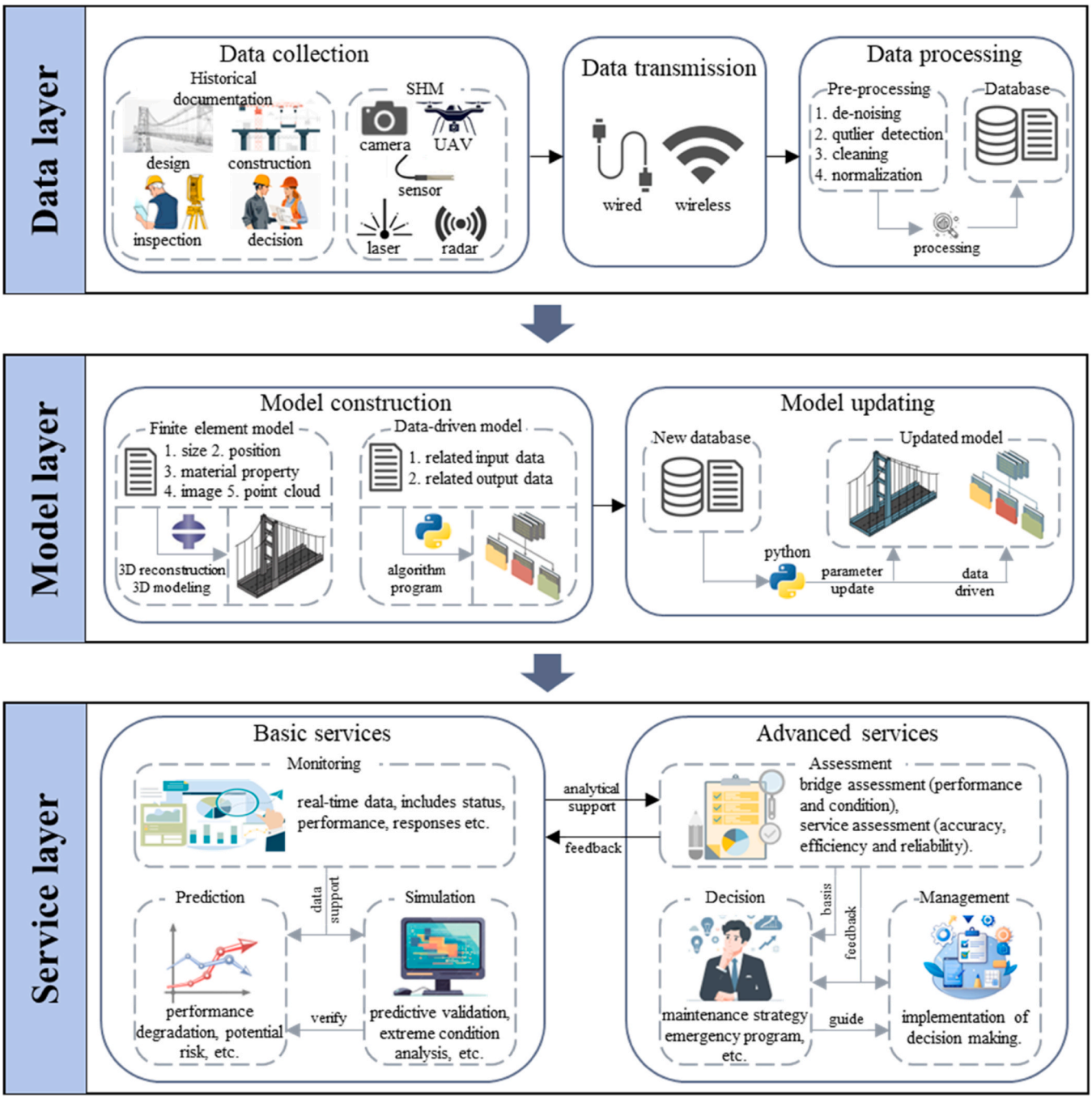


Fig. 6. Three-layer framework for DT in health monitoring and condition assessment of cable-supported bridge.

model can effectively reduce modeling workload and offer higher accuracy and broader adaptability. A key characteristic of DT is the real-time connection and data transmission between the virtual model and the physical entity, which requires the virtual model to be updated in real-time. This framework provides an approach for updating the virtual model, which is based on the real-time updated database and achieves automated updating of finite element model parameters and data-driven model parameters through Python scripts. Specifically, Python scripts are used to read data from the database in real-time and automatically modify the input file (INP file) of the Abaqus model to achieve real-time updating of the finite element model. Python scripts are also employed for recalculating and optimizing parameters in the data-driven model to achieve real-time updating.

The service layer refers to the services provided by the DT to the physical entity, including monitoring, prediction, simulation, assessment, decision and management. These six services are all crucial for health monitoring and condition assessment of cable-supported bridges. They can be categorized into basic services and advanced services. The basic services include monitoring, prediction, and simulation, which provide essential data and analysis support for understanding the current and future behavior of the bridge. The advanced services include assessment, decision making, and management, which build on the basic services to support complex operational and management activities.

It is worth noting that the three-layer digital twin framework presented a more logically structured and detailed interpretation compared to the service description in Section 4.4. Monitoring service provides real-time insight into the performance and condition of the bridge. Prediction service analyzes the future performance degradation or

potential risk of the bridge based on the monitoring service, and simulation service verifies the results provided by the prediction service or analyzes the structural response under extreme conditions. Assessment service integrates the three services included in the basic service, not only assessing the condition of the bridge, but also evaluating and analyzing the quality and effect of other services, achieving feedback and optimization. Decision service generates maintenance strategies or emergency plans based on the results of the assessment service. Management service implements the decision to ensure efficient operation of the bridge. Operators of cable-supported bridges can prioritize certain services based on actual needs, allowing the DT to be tailored to specific operational and managerial requirements.

7. Challenges and future trends

7.1. Challenges

The application of DT may encounter some challenges. According to the results of the literature review in this paper, the following challenges may be faced.

In terms of data management, higher-precision and durable sensors are required for data collection and data loss should be minimized. When collecting huge amounts of data, low latency and high reliability must be ensured during data transmission and processing.

In terms of modeling, the real-time performance and fidelity of existing virtual models are not outstanding. Virtual models need to reflect physical entities in a more real-time and high-fidelity manner. Additionally, model updating and fusion require further enhancement.

In terms of security and privacy, while DT may need to be connected to a network to exhibit its excellent characteristics, a large amount of data may face the risks of network threats and attacks, as well as data theft or tampering, necessitating strengthened supervision and control.

In terms of service applications, the application of DT in cable-supported bridges is still limited, and more engineering practice and testing are needed to better serve health monitoring, condition assessment, and early warning.

In terms of intelligence, scholars and stakeholders expect DT to be a system that can autonomously, adaptively, and self-evolvingly perform condition assessment, prediction, and decision-making based on collected data. However, current DT still requires human intervention, and advanced technologies need to further enhance its intelligence.

In terms of integration, DT involves the integration of multiple technologies and disciplines, and how to efficiently and fully integrate these systems remains a challenge.

In terms of standardization, there is currently a lack of unified standards in the bridge field to guide the construction of DT systems and the evaluation of model quality.

7.2. Future trends

Through the review and discussion of relevant research on the application of DT, it is found that the ultimate goal is to construct a DT system for the physical entity of cable-supported bridge. This system is characterized by real-time perception and access to the entire scenario, high-fidelity reproduction of the physical entity, and simultaneous intelligent provision of multiple services, providing critical assistance and problem-solving solutions for decision-makers. However, this goal still faces numerous challenges. To this end, this paper proposes several suggestions:

(1) Fusion of multiple technologies in DT. It is worth noting that the future development of DT will primarily focus on digitization, integration, and intelligence. The combined application of visual reality, augmented reality technologies and DT can significantly enhance the visualization and interactivity, which in turn improves the efficiency and accuracy of bridge monitoring and assessment.
(2) Develop standards for the construction of bridge DT systems to standardize the core elements and minimum indicator requirements. This will play a significant role in advancing the development of DT in this field.
(3) Establish a clear evaluation system for the performance of bridge DT, which can quantitatively and rapidly assess the performance of a DT. The evaluation system can be constructed from perspectives such as accuracy, efficiency, multi-service capability, ease of deployment, and intelligence level. The evaluation system can horizontally evaluate different DTs to select outstanding ones and also guide the direction of DT development.

8. Conclusions

In recent years, DT has received significant attention due to its exceptional performance and has achieved notable progress in health monitoring and condition assessment of cable-supported bridges. The emergence of DT has compensated for the deficiencies of traditional methods that often cannot deal with scattered data, but it also faces some challenges.

This paper provides a systematic review of the research and application of DT in health monitoring and condition assessment of cable-supported bridges.

It starts by introducing the origin, definition, and characteristics of DT, and provides a specific definition of DT in the field of health monitoring and condition assessment of cable-supported bridges. Additionally, a comparison between DT and similar concepts is presented.

Subsequently, the current research status is analyzed through a systematic literature review with a bibliometric analysis approach. The construction methods of the DT system are then discussed regarding data collection, data transmission, DT modeling, and services provided by DT. Furthermore, we propose a three-layer framework of DT tailored for health monitoring and condition assessment of cable-supported bridges. Finally, the challenges and future trends are discussed.

CRediT authorship contribution statement

Zengpeng Zhang: Writing – original draft, Investigation, Formal analysis, Data curation. **Yuguang Fu:** Validation, Investigation, Formal analysis. **Zhen Sun:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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